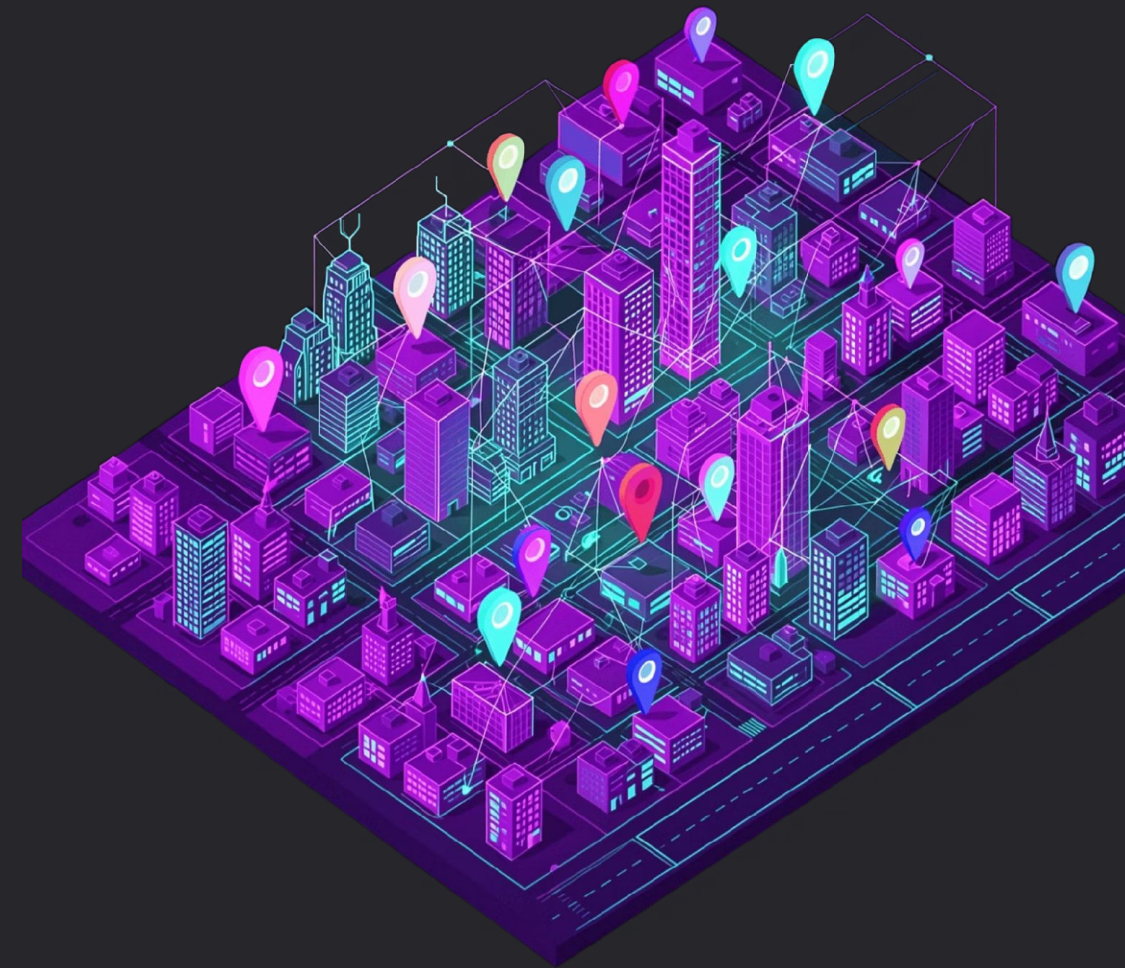


Intent-Driven Spatial Search

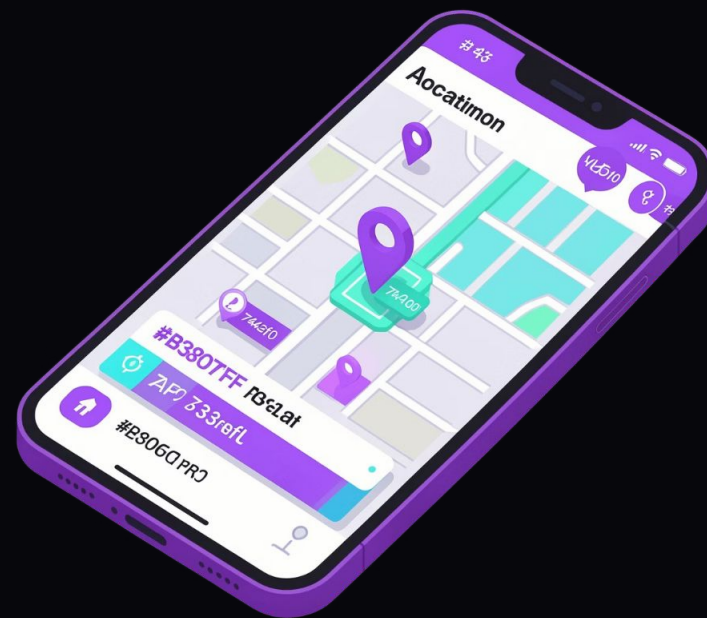
A Hybrid NL-to-SQL Architecture for Conversational Geographic Queries

Replacing catastrophic keyword failures with intent-aware, executable PostGIS routing — evaluated on a 50-query stratified benchmark for the Pilani region.



THE PROBLEM

Catastrophic Spatial Prioritization Failure



When a user in Pilani searches "**quiet places near me**," mainstream engines abandon the spatial constraint entirely — surfacing a venue named "Quiet Place" in **Bengaluru**, over **1,000 km** away.



Root Cause: Deterministic keyword filtering matches business-name indexes, not semantic intent. The explicit "near me" constraint is silently discarded in favor of textual relevance.

In India, where spatial references are informal and context-rich, this architectural gap renders digital maps effectively useless for intuitive local exploration.

Three Fundamental Architectural Failures

Exact-Match Bias

Systems prioritize exact word matches in business titles over geographic proximity — a venue *named* "Quiet" outranks a genuinely quiet park 200 m away.

Deterministic Filtering

Colloquial descriptors ("uncrowded," "safe," "study spot") cannot be dynamically mapped to formal geographic tags. No semantic bridge exists between intent and schema.

Decoupled Architecture

NLP and the spatial database operate in isolation. Spatial constraints are abandoned in favor of semantic text matching, breaking the locality guarantee.

From Natural Language to Executable PostGIS SQL



User Query

Semantic Intent

Defog SQLCoder

PostGIS DB

The architecture eliminates cloud API dependency in its offline configuration, routing conversational vocabulary directly into executable, strictly bounded spatial commands — semantic intent is natively understood at the data layer.

DATA LAYER

Geographic Ground Truth & Semantic Intent Ontology

PostgreSQL / PostGIS Database

A denormalized `pilani_places` table from OpenStreetMap, intentionally tag-heavy and chaotic. Schema groups: core identifiers, PostGIS geometry columns, place categories, and utility tags (`opening_hours`, `payment:upi`).

Semantic Intent Ontology

A structured mapping dictionary bridging colloquial vocabulary to schema fields:

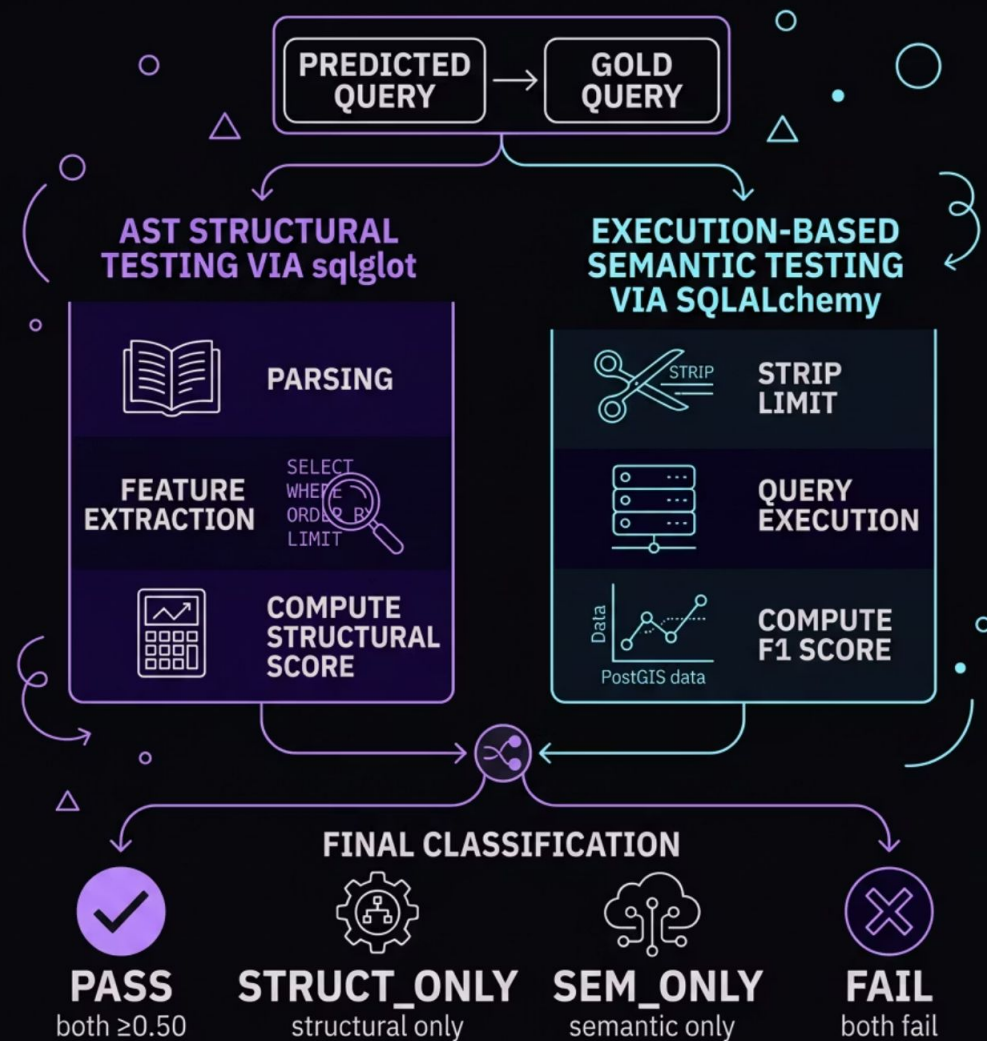
Late Night → `opening_hours ILIKE '%24/7%'`

Study → `amenity ILIKE '%library%'`

Digital Payment → `payment:upi = 'yes'`

Hybrid Evaluation Framework

Academic baselines fail on spatial schemas: Spider produces **false negatives** (penalizing valid alternative SQL), while Test Suite produces **false positives** (accidental matches from data sparsity). The proposed framework fuses both.



Golden Benchmark Suite

50 hand-authored queries stratified by difficulty (Easy / Medium / Hard / Extra), each with a manually verified Gold SQL. Queries span Food, Academics, Residential, and Logistics intent clusters.



Pass threshold: structural score ≥ 0.50 **and** semantic F1 ≥ 0.50 — both conditions must be satisfied.

Configuration A: Cloud-Served SQLCoder-8B

84%

Overall Pass Rate

42 of 50 queries passed both structural and semantic thresholds

0.827

Avg. Structural Score

Strong schema linking — correct tables, columns, and operators

0.784

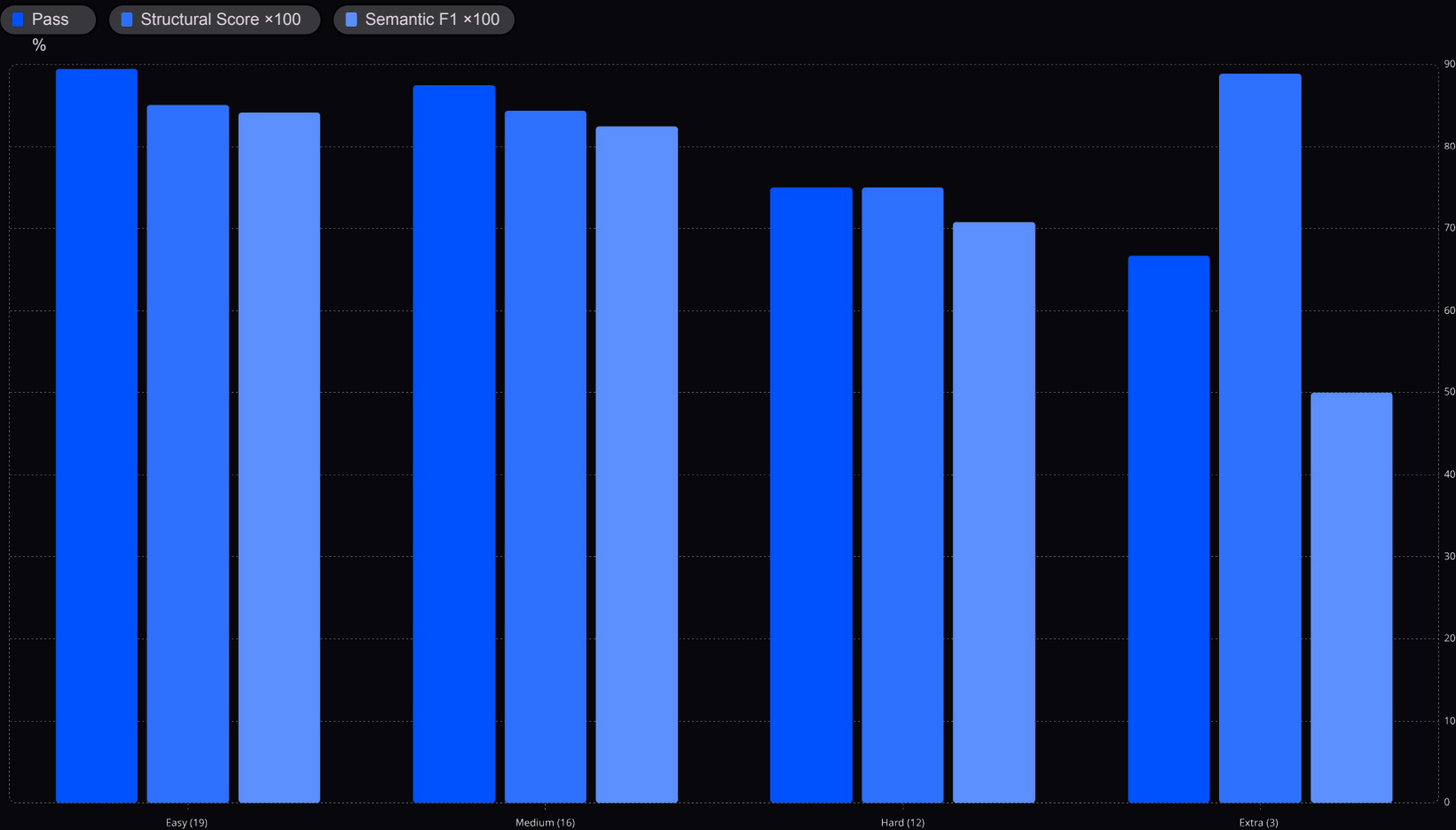
Avg. Semantic F1

High entity-set overlap against live PostGIS execution

50

Benchmark Queries

Stratified across 4 difficulty tiers and 4 intent clusters



Performance degrades gracefully. Extra Hard queries show a structural rebound (0.889) but F1 lands exactly at the 0.500 threshold — well-formed SQL that misses the full semantic subset.

Configuration B: llama.cpp + 4-bit GGUF + Kimi 2B Parser

Motivation

- Eliminate API latency and per-token cost
- Ensure data privacy without campus Wi-Fi
- Run entirely on local CPU/GPU with reduced VRAM

Observed Regressions

INT4 quantization introduced precision loss in schema-linking — near-correct column names and malformed JOINS on Hard/Extra queries
Kimi 2B parser produced under-specified prompts, amplifying the partial-match problem from Config A



**CONFIGURATION A
(CLOUD
SQLCODER-8B):**
**84% Pass Rate,
Structural 0.827,
Semantic F1 0.784,
Full Grammar Parser,
NO Latency, Cloud
Dependency**



**CONFIGURATION B
(OFFLINE LLAMA.CPP
4-BIT + KIMI 2B):**
**Accuracy Regression on
Hard and Extra Tiers,
INT4 Precision Loss in
Schema-Linking,
Weaker Grammatical
Decomposition, ZERO
Cloud Dependency,
FULL Data Privacy,
Local CPU/GPU Only**

Two Architectural Limitations

Over-Aggressive Logic Grouping

When users ask for *"volleyball or table-tennis courts,"* the model applies **AND** logic — searching for venues with both tags simultaneously. This artificially narrows the bounding box, dragging F1 below 0.50. **14% of queries** classified as STRUCT_ONLY.

String-Matching Fragility

Entity resolution for unregistered colloquial nicknames remains incomplete. A search for *"Meera Bhawan girls hostel"* fails when the database lists only *"Meera Bhawan."* The ontology resolves generic categories but not building-specific aliases.

- ✅ Despite edge cases, intent-based NL-to-SQL routing achieves **84.0% pass rate** — a decisive improvement over flat keyword indexing for localized spatial search.

From Prototype to Production System

Work Completed

- PostgreSQL/PostGIS database + Semantic Intent Ontology for Pilani
- 50-query Golden Benchmark Suite across 4 intent clusters
- Hybrid Evaluation Framework: 84% pass rate, $F1 \geq 0.50$
- Functional offline prototype via llama.cpp 4-bit GGUF

Future Work Agenda

- Replace Kimi 2B with a 7B-class on-device conversational front end
- Refine prompts to enforce OR-grouping for disjunctive intents
- Expand ontology with localized building aliases and hostel nicknames